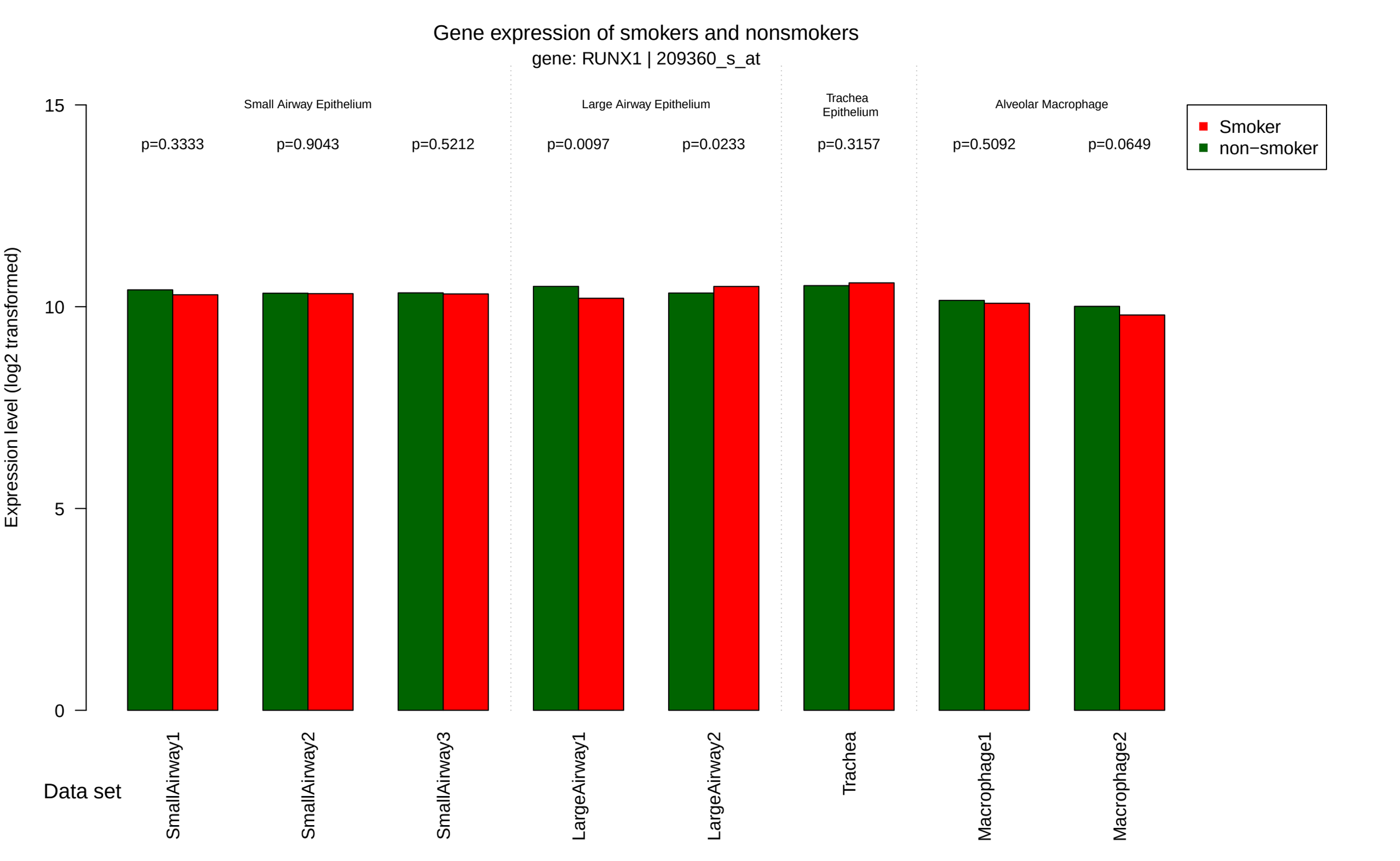
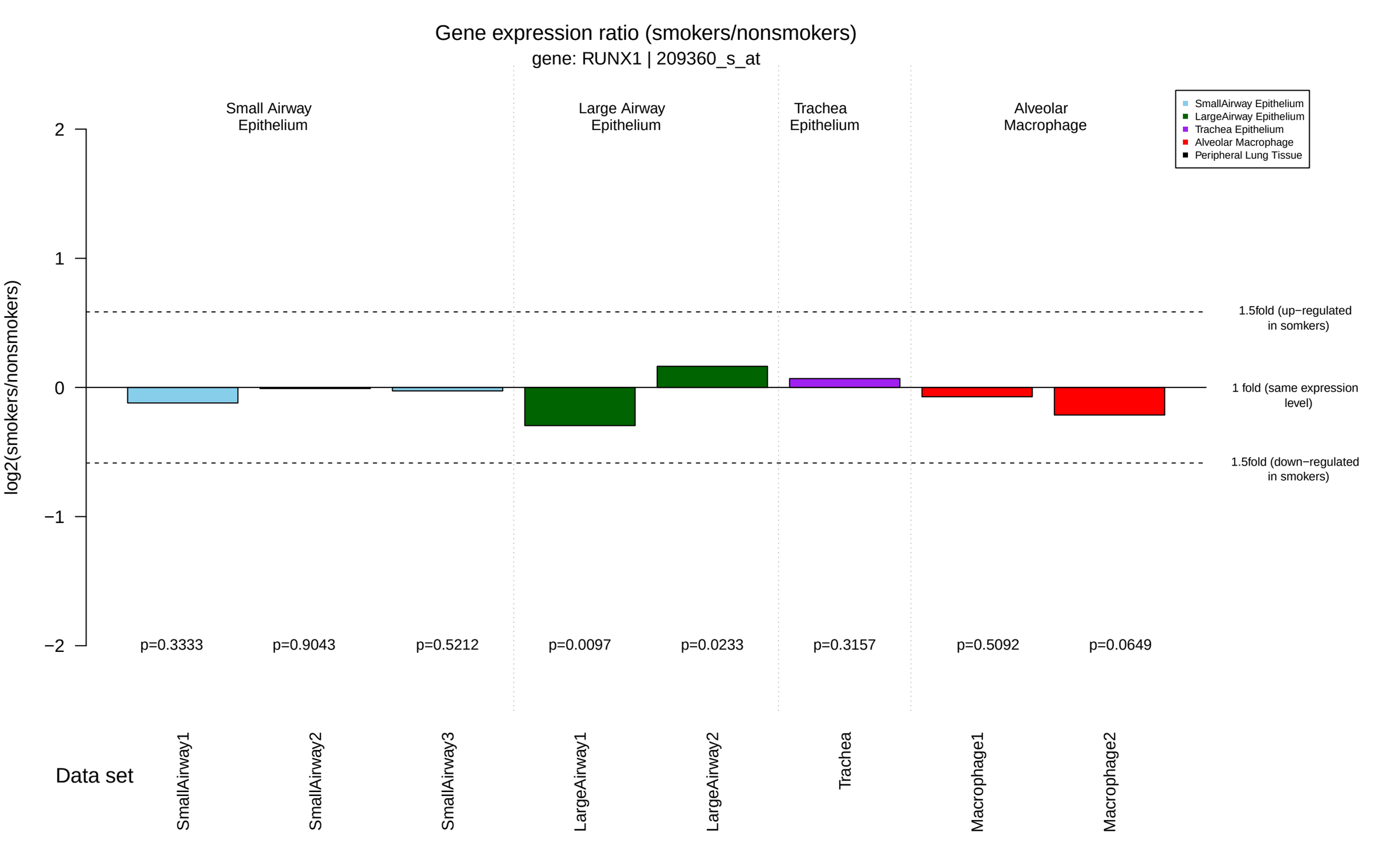
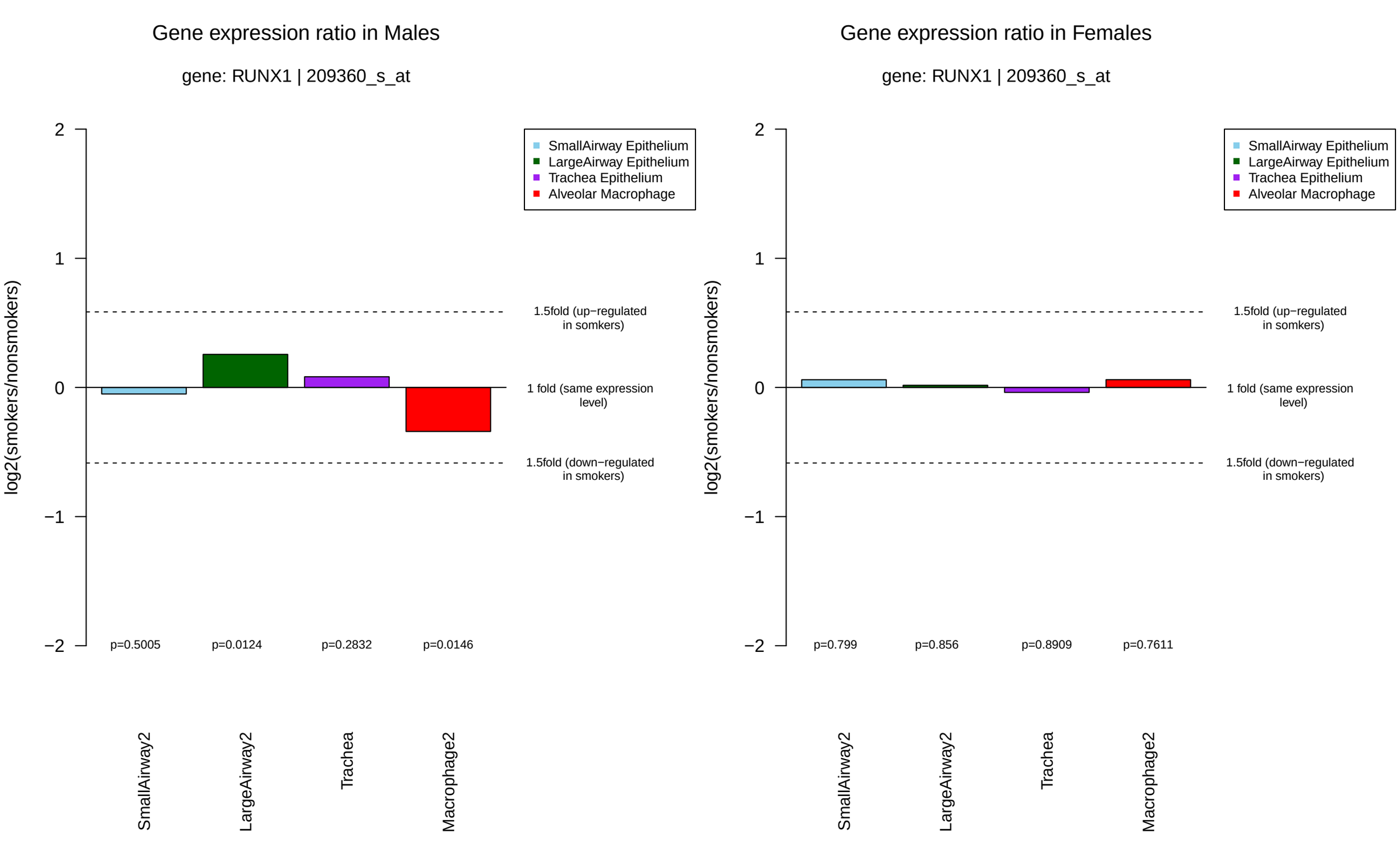
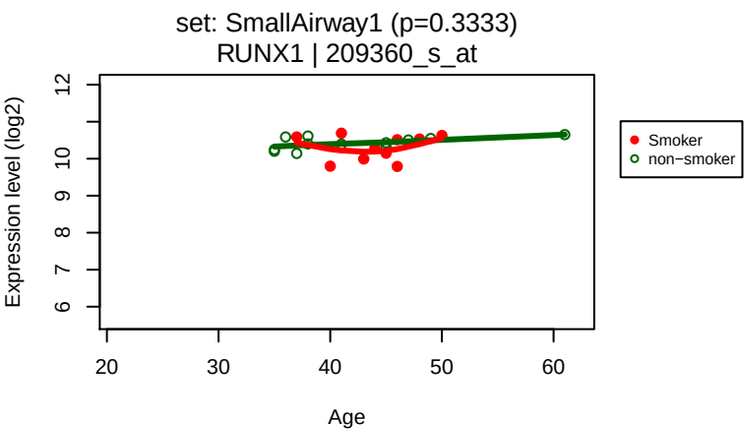
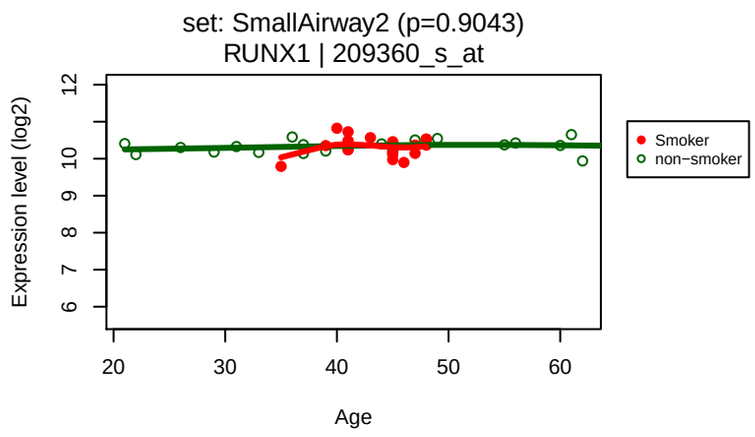
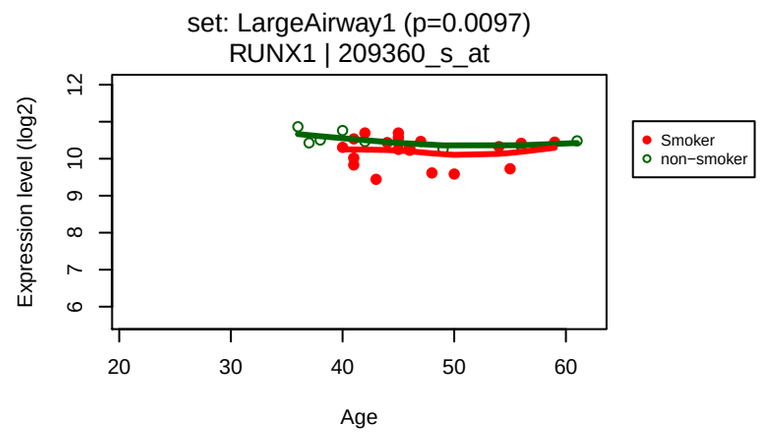

Gene expression of smokers and nonsmokers
gene: RUNX1 | 209360_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3333
p=0.9043
p=0.5212
p=0.0097
p=0.0233
p=0.3157
p=0.5092
p=0.0649
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

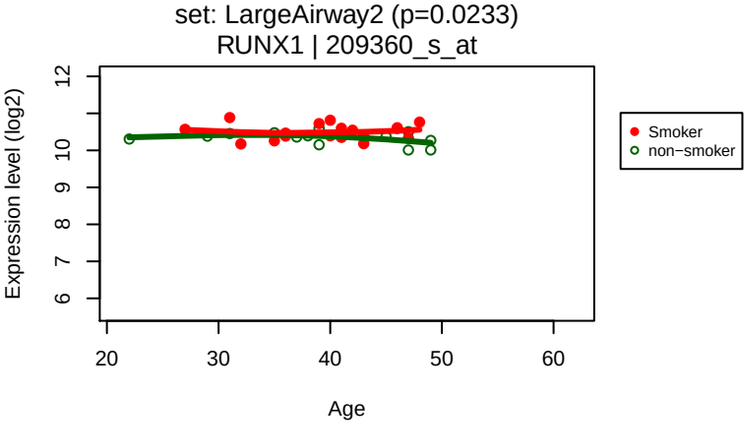

Gene expression ratio (smokers/nonsmokers)
gene: RUNX1 | 209360_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3333
p=0.9043
p=0.5212
p=0.0097
p=0.0233
p=0.3157
p=0.5092
p=0.0649
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

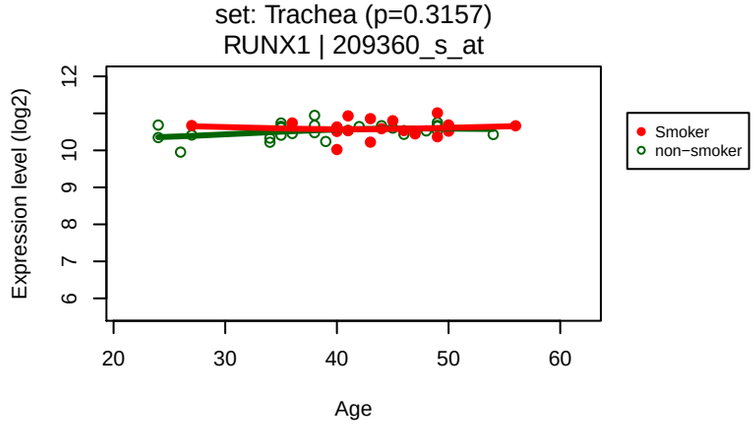

Gene expression ratio in Males
gene: RUNX1 | 209360_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5005
p=0.0124
p=0.2832
p=0.0146
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RUNX1 | 209360_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.799
p=0.856
p=0.8909
p=0.7611
SmallAirway2
LargeAirway2
Trachea
Macrophage2

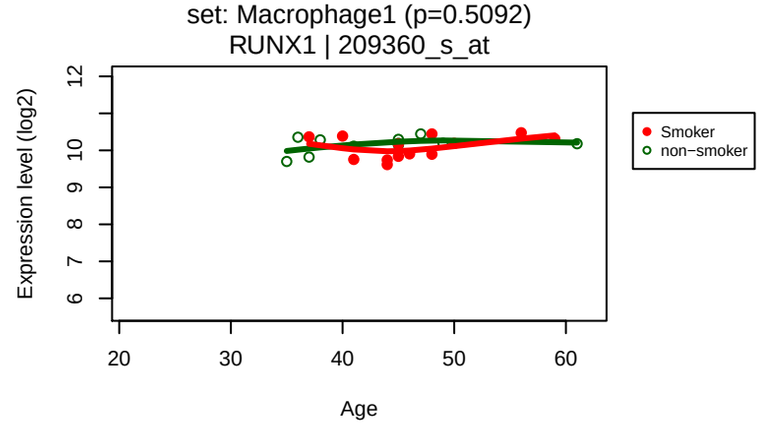

set: SmallAirway1 (p=0.3333)
RUNX1 | 209360_s_at
Expression level (log2)
Age
Smoker
non−smoker

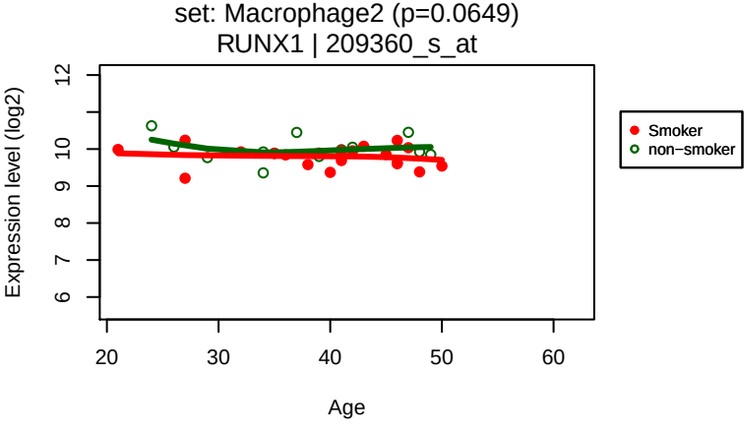

set: SmallAirway2 (p=0.9043)
RUNX1 | 209360_s_at
Expression level (log2)
Age
Smoker
non−smoker

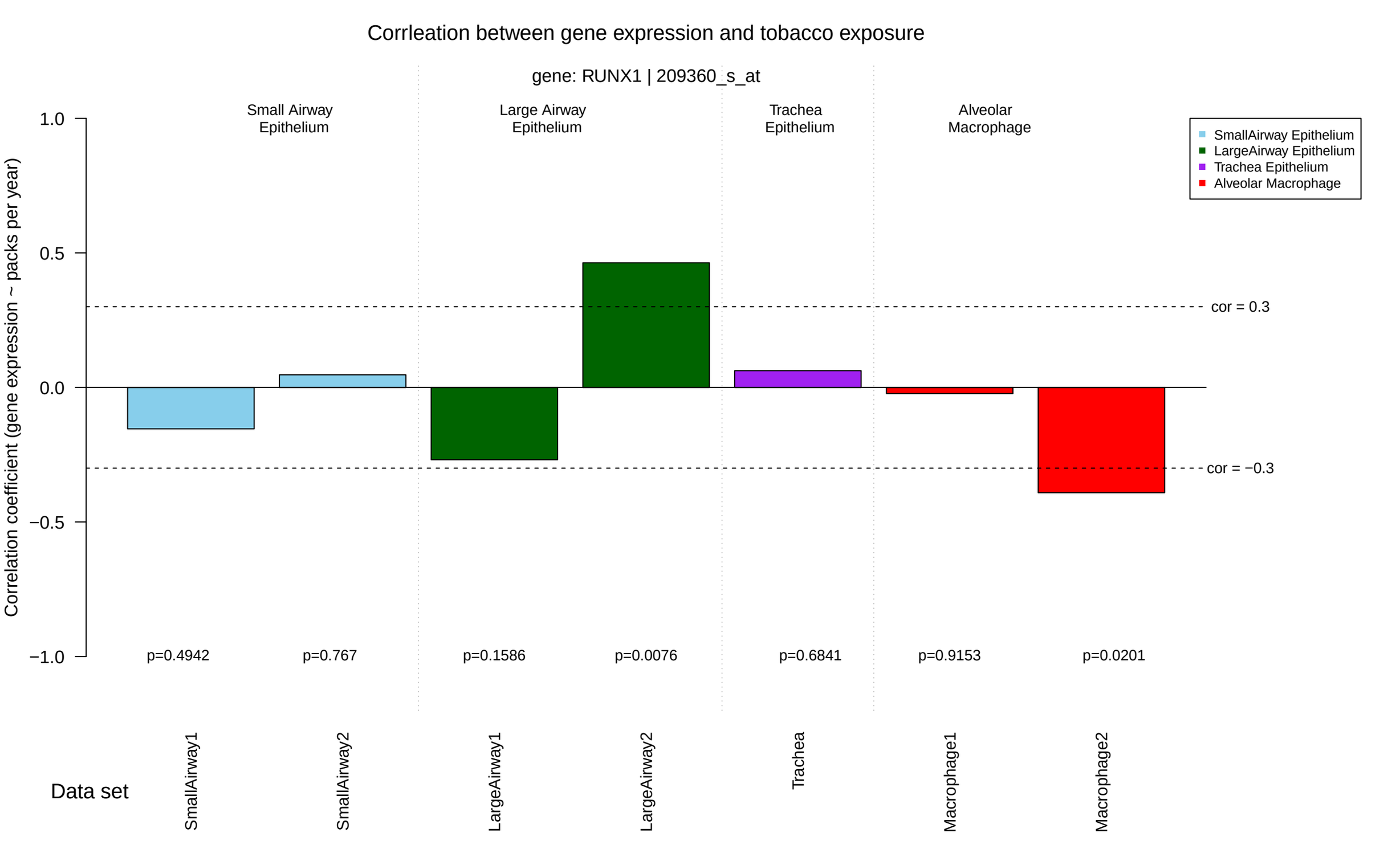

set: LargeAirway1 (p=0.0097)
RUNX1 | 209360_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0233)
RUNX1 | 209360_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3157)
RUNX1 | 209360_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5092)
RUNX1 | 209360_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0649)
RUNX1 | 209360_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RUNX1 | 209360_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4942
p=0.767
p=0.1586
p=0.0076
p=0.6841
p=0.9153
p=0.0201
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2